

CUSTOMER SALES ANALYSIS REPORT

Executive Summary

This project analyzes customer purchasing behavior by integrating sales transaction data with customer churn information. The objective is to identify high-value customers, sales trends, regional performance, churn patterns, and cross-selling opportunities.

Using Python (pandas, seaborn, matplotlib), the analysis performs data cleaning, feature engineering, aggregation, pivot tables, and visualization to generate actionable business insights.

1. Key Performance Indicators

- **Total Revenue:** Calculated from all completed sales transactions
- **Total Customers:** Unique active customers in the dataset
- **Average Order Value:** Average revenue per transaction
- **Top Customer:** Highest Lifetime Value contributor
- **Customer Churn Rate:** Percentage of customers who discontinued service

These KPIs provide a snapshot of overall business health.

2. Data Processing Overview

- Converted Date fields to datetime format
- Normalized customer IDs across datasets
- Handled missing values in financial fields using median imputation
- Created derived features such as:
 - Monthly sales periods
 - Customer Lifetime Value

Sales and churn datasets were merged to enable combined behavioral analysis.

3. Sales Performance Insights

3.1. Regional Performance

The North region generates the highest revenue, followed closely by the South region. The East region shows moderate performance, while the West region records the lowest revenue contribution

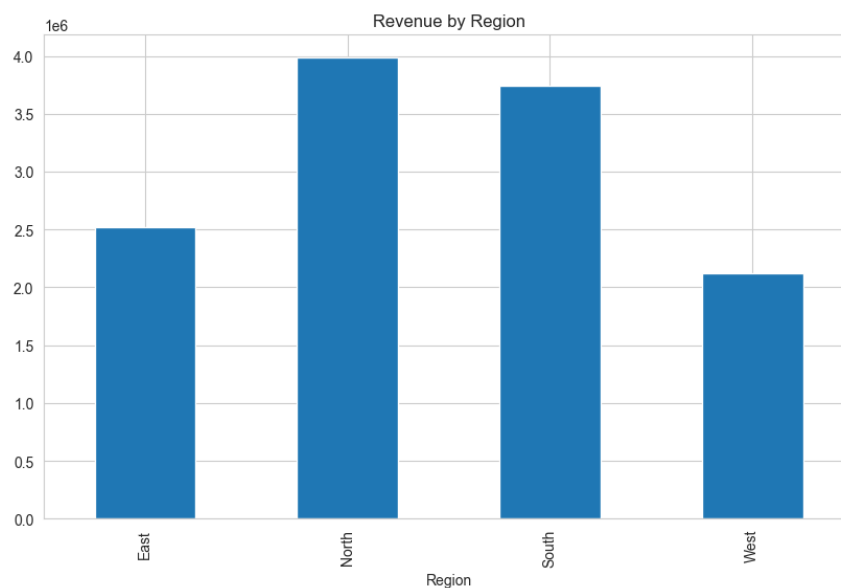


Fig.3.1.1. Revenue by Region

This indicates that revenue is concentrated primarily in North and South, suggesting stronger customer demand or higher purchasing power in these regions.

Insight:

North and South act as primary revenue drivers, while West presents an opportunity for growth.

3.2. Product Performance

Among all products, Laptop is the highest revenue-generating product, followed by Tablet and Phone. Headphones and Monitor contribute comparatively lower revenue.

Revenue is therefore driven mainly by premium devices, with accessories playing a secondary role.

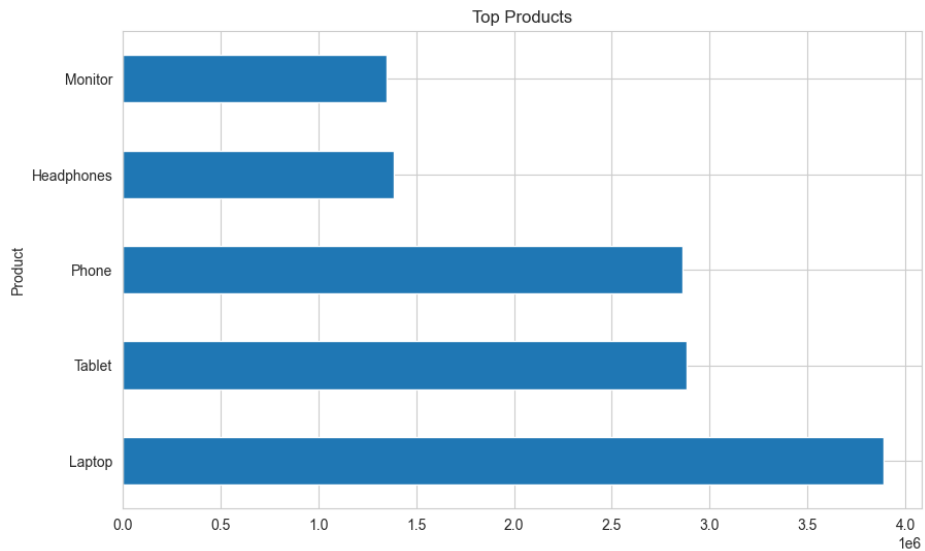


Fig.3.2.1. Top Product

Insight:

Sales are heavily dependent on a small set of high-value products.

3.3. Seasonal Trends

Monthly sales analysis shows noticeable fluctuations:

- Sales peak in **March**.
- A decline is observed in **February**.
- April shows a sharp drop, indicating reduced activity or possible seasonality.

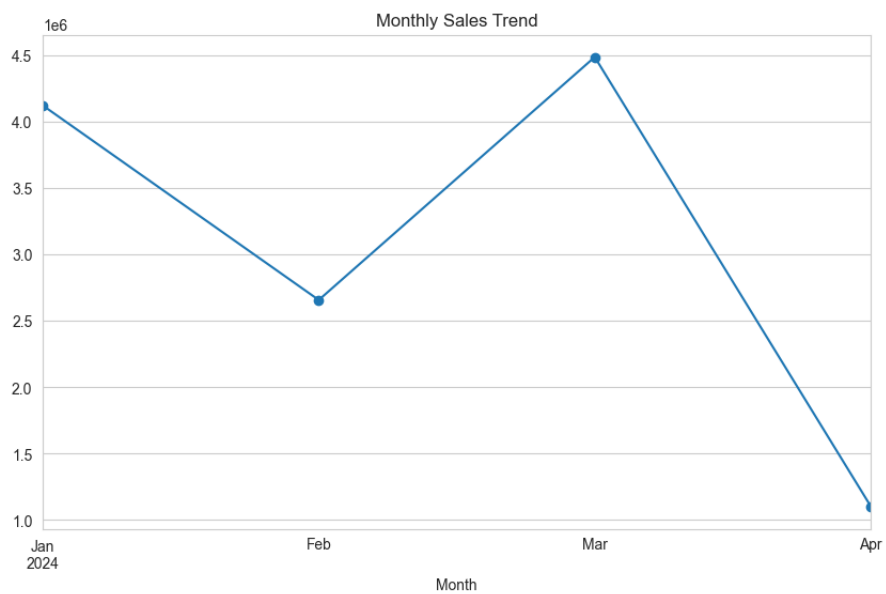


Fig.3.3.1. Monthly Sales Trend

Insight:

Sales exhibit clear month-to-month variability, suggesting seasonal demand patterns.

High-performing months can be leveraged for promotions, while low-performing months require customer engagement strategies.

4. Customer Value Analysis

Customer Lifetime Value distribution reveals:

- A majority of customers fall into the low-to-mid value range.
- A small group of customers contributes exceptionally high lifetime revenue.

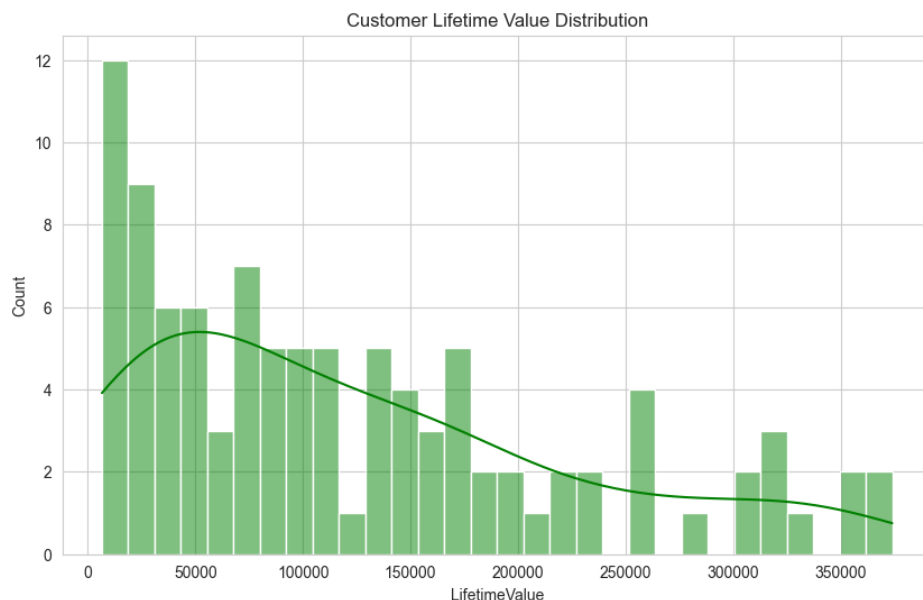


Fig.4.1. Customer Lifetime Value Distribution

This reflects a revenue concentration pattern where a limited number of customers drive a large share of total sales.

5. Retention Analysis

Churn distribution shows that most customers are retained; however, a measurable portion has churned.

Although churn is not dominant, even a small churn percentage represents potential revenue loss, especially when high-value customers are involved.

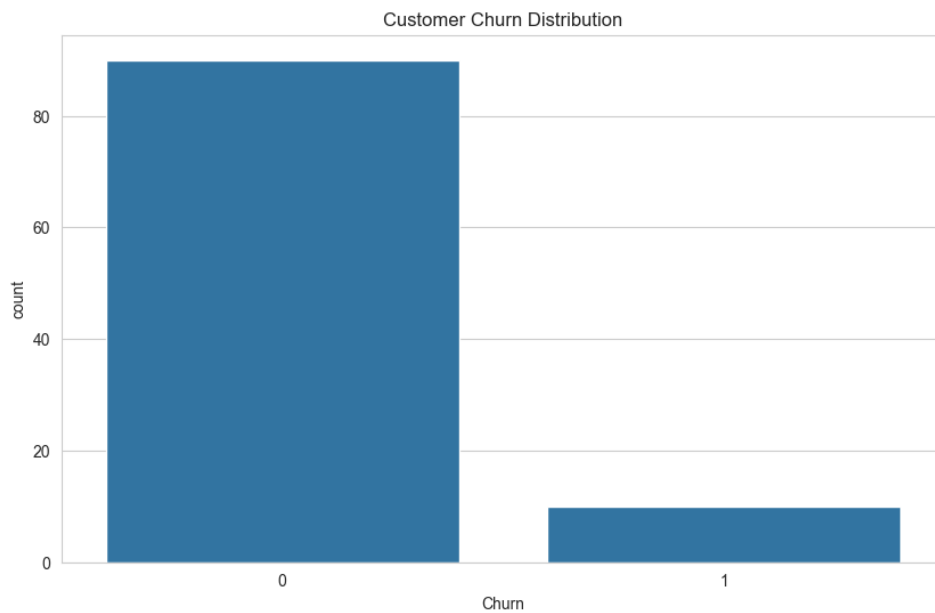


Fig.5.1. Customer Churn Distribution

Insight:

Targeted retention efforts are needed to minimize churn and stabilize long-term revenue.

6. Cross-Selling Opportunities

Product correlation analysis indicates weak and mostly negative relationships across all product pairs.

This suggests that customers generally purchase single products per transaction, and no strong natural cross-selling patterns exist in the current dataset.

	Product	Headphones	Laptop	Monitor	Phone	Tablet
Product						
Headphones		1.000000	-0.236067	-0.176471	-0.210042	-0.249004
Laptop		-0.236067	1.000000	-0.236067	-0.280976	-0.333096
Monitor		-0.176471	-0.236067	1.000000	-0.210042	-0.249004
Phone		-0.210042	-0.280976	-0.210042	1.000000	-0.296374
Tablet		-0.249004	-0.333096	-0.249004	-0.296374	1.000000

Fig. 6.1. Correlation Matrix for Products Bought Together

Insight:

Organic cross-selling behavior is minimal.

Strategic Recommendations

1. Strengthen retention for high lifetime value customers.
2. Expand presence in North and South regions.
3. Improve West region performance through promotions.
4. Introduce bundled product offerings.
5. Use monthly trends for demand forecasting.

Conclusion

The analysis identified key revenue-driving regions and products while highlighting customer value concentration and limited cross-selling behavior. Strategic focus on high-performing regions, high-value customers, and bundled offerings can significantly improve revenue growth and customer retention.